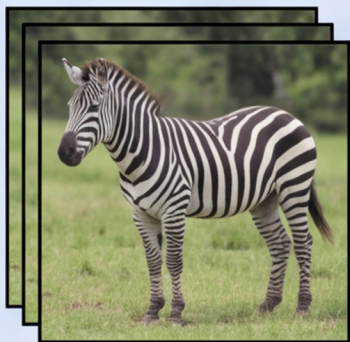
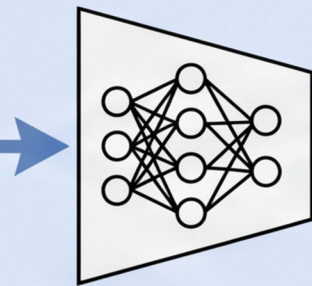


"Phase 1: Neural-to-semantic transition"

Input representations and latent matrix

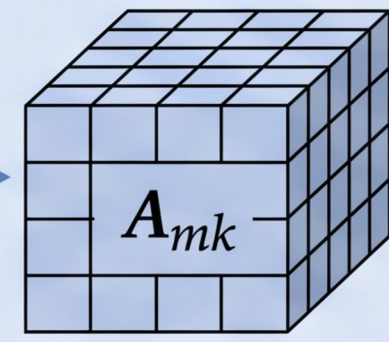


Training image (AwA2) benchmark



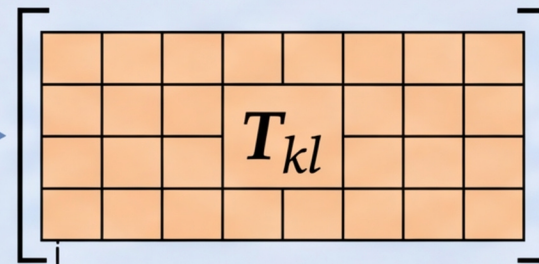
Frozen ResNet-101 feature extractor

Uninterpretable latent feature space



m = training dataset universe objects
 k = 2048-dimensional neural feature

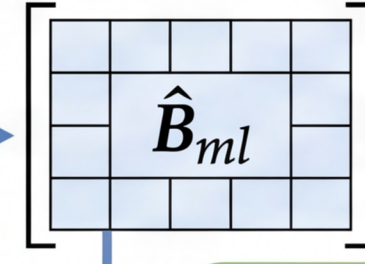
Global transition operator



Rank- r SVD basis V_r

Auxiliary principal component PA compresses collinear neural features apply linear ridge regression weights

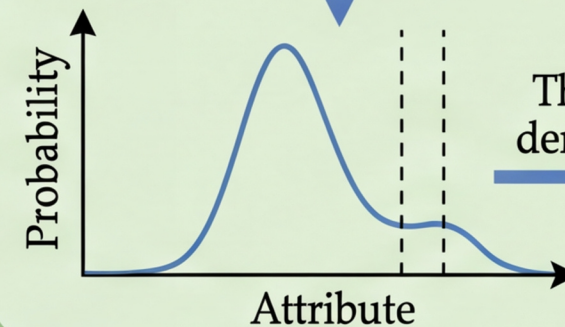
Reconstructed continuous semantic attributes



$$\hat{B} = (A - \bar{1})^T T + \bar{1} \cdot b_0^T$$

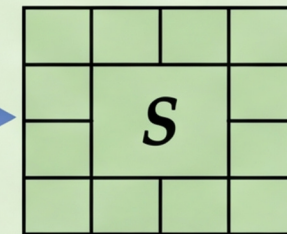
Phase 2.1: Semantic attribute selection and stable discretization (WEDD)

Physical conceptual salience



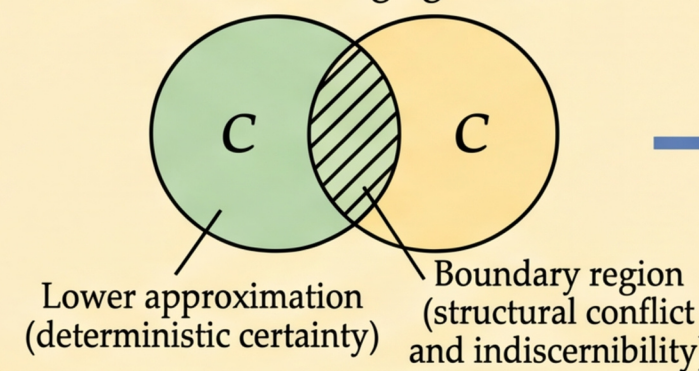
Threshold density cuts

Discrete symbolic states



Phase 2.2: Rough-set granulation and knowledge discovery

Formal knowledge granulation



Global conflict-aware production rulebook $\mathcal{R}$

$$r : \bigwedge_{j \in R_r} [s_j = v_j] \Rightarrow d = y_r$$

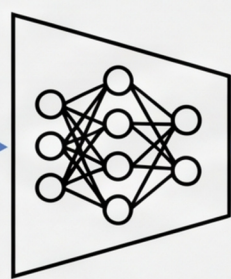
IF paws = s2 AND hunter = s2 AND small = s0
AND yellow = s2 $\Rightarrow$
class: tiger (confidence 0.896)

Phase 2.3: Conflict-aware inference and evaluation

Static rules induced audit real-time unobserved data



Test image



Frozen ResNet-101

$a \times k$
unseen test feature vector

Project via transition weights
 T

$\hat{b} \times l$
unseen continuous semantic

Apply WEDD density cuts

s
active test symbolic signature

Query gate
Global rulebook $\mathcal{R}$



Exact match: Covered fidelity and covered accuracy (verifiable logical subset)



Multiple rule firing: Resolve via integral support score $S(y_k)$



Structural gap: Soft match via Hamming distance d_H or trigger audit abstention